

Introduction to Algebra (II) HW5

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1

Let ζ_n be a primitive n -th root of unity and $K_n = \mathbb{Q}(\zeta_n)$. Since the minimal polynomial of ζ_n over $\mathbb{Q}$ is the cyclotomic polynomial $\Phi(x)$, which has degree $\varphi(n)$.

$\mathbb{Q}(\zeta_n)$ contains all roots of $\Phi(x)$, so it's a splitting field, hence normal. Since $\mathbb{Q}$ is characteristic 0 field, $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is separable. Here we have $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois.

$G = \text{Aut}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ acts on the roots of $\Phi(x)$ transitively, and every automorphism $\sigma \in G$ is determined by its action on ζ_n . Since $\sigma(\zeta_n)$ must again be a primitive n -th root of unity, we have:

$$\sigma(\zeta_n) = \zeta_n^k \text{ for some } k \in (\mathbb{Z}/n\mathbb{Z})^\times$$

This gives a well-defined injective homomorphism:

$$\pi : \text{Aut}(K_n/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times, \sigma_k \mapsto k$$

where σ maps ζ_n to ζ_n^k for some $k \in (\mathbb{Z}/n\mathbb{Z})^\times$ (The homomorphism is easy to check. $\pi(\sigma_a \circ \sigma_b) = \pi(\sigma_{ab}) = ab = \pi(\sigma_a)\pi(\sigma_b)$, $\circ$ stands for composition here)

Because both $\text{Aut}(K_n/\mathbb{Q})$ and $(\mathbb{Z}/n\mathbb{Z})^\times$ has the order of $\phi(n)$, this is an isomorphism.

2

Let $f(x) = x^{2^n} + x + 1$ be defined over $\mathbb{F}$. Assume it's irreducible.

Let α be a root of $f(x)$ in the splitting field of f , by definition we have:

$$\begin{aligned}\alpha^{2^n} + \alpha + 1 &= 0 \\ \alpha^{2^n} &= -\alpha - 1 = \alpha + 1 \\ (\alpha^{2^n})^{2^n} &= (\alpha + 1)^{2^n} \\ \alpha^{2^{2n}} &= \alpha^{2^n} + 1 = -\alpha = \alpha\end{aligned}$$

The relation $\alpha^{2^{2n}} = \alpha$ guarantees that α is a element of $\mathbb{F}_{2^{2n}}$.

$[\mathbb{F}_{2^{2n}} : \mathbb{F}_2] = 2n$, since we assume that f is irreducible, the degree of f must divide $2n$:

$$2^n \mid 2n$$

When $n = 1, 2$, the relation holds. But when $n > 3$, $2^n > 2n$ is always true (This can be proved by induction), so $2^n \nmid 2n$ when $n > 3$. In conclusion, f is irreducible only when $n = 1$ or $n = 2$.